

HealthConnect Clinic: Advanced Analytics & Validation Report

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1. Executive Summary & Baseline Validation

Building directly upon the 5,000-record baseline dataset established in Week 5 (overall baseline no-show rate of 48.46%), Week 6 transitions from exploratory analysis into formal statistical hypothesis validation and non-linear feature engineering. We formally evaluated the primary drivers of appointment attendance—specifically booking lead times, clinic queue waiting times, and reminder notification channels—to confirm whether observed variations represent genuine operational patterns or random sample noise.

2. Statistical Hypothesis Validation Matrix

Rigorous statistical testing was performed across key operational variables using Chi-Square Tests of Independence and Two-Sample Independent T-Tests. All tested drivers demonstrated statistical significance at the $p < 0.001$ threshold:

Tested Operational Driver	Statistical Test	Metric / Test Stat	p-Value	Validation Outcome & Interpretation
Reminder Communication Impact	Chi-Square (χ^2)	$\chi^2 = 27.52$	1.56×10^{-7}	Statistically Significant. Confirming SMS/WhatsApp reminders reliably improve appointment compliance.
Booking Lead Time Duration	Two-Sample t-test	$t = -15.87$	2.31×10^{-55}	Statistically Significant. Longer lead times strongly correlate with increased patient drop-off.
Clinic Queue Waiting Time	Two-Sample t-test	$t = -12.82$	5.06×10^{-37}	Statistically Significant. Excessive wait times (>30 mins) drive attendance attrition.

3. Non-Linear Segment Risk Analysis

Deeper cross-tabulation across patient cohorts revealed critical non-linear behavior patterns:

- The 21-Day Lead Time Risk Cliff:** No-show probability remains manageable (~36%–42%) for bookings scheduled within 0 to 7 days. However, when lead times cross 21 days, no-show rates jump sharply to over **63%**, identifying long lead times as HealthConnect's primary revenue leak.
- Operational Bottlenecks:** Queue wait times exceeding 30 minutes cause severe drop-off rates, disproportionately affecting senior age cohorts (56+) who face greater mobility and scheduling constraints.

4. Data Science Cross-Track Handoff

Data Analytics → Data Science Interface: To resolve high false-negative classification errors in the predictive baseline model, the Data Analytics track engineered non-linear binned features and exported the artifact `HealthConnect_Feature_Engineered_Segments.csv`.

- **Engineered Feature 1** (`lead_time_bracket`): Segmented into 0-7 Days, 8-21 Days, 22-45 Days, and 45+ Days.
- **Engineered Feature 2** (`age_group`): Categorized into 18-35, 36-55, and 56+ cohorts.
- **Engineered Feature 3** (`high_wait_flag`): Binary flag (`1` if wait time > 30 minutes, `0` otherwise).

5. Actionable Strategic Recommendations

- **Automated Reminder Sequences:** Implement mandatory 48-hour automated SMS/WhatsApp alerts, targeting long-lead bookings.
- **Overbooking Buffers:** Apply dynamic overbooking caps exclusively to booking slots scheduled beyond 21 days out.
- **Queue Fast-Tracking:** Establish fast-track dispatch for senior patients when clinic queue wait times reach 25 minutes.